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METHOD FOR CONSOLIDATING MULTILAYER FOILS USING HOT PRESSING

Abstract

A method for consolidating a foil tab stack is provided. The method includes combining foil tabs of an electrode stack in a battery cell to form a combined foil tab stack. Additionally, the method includes hot-pressing the combined foil tab stack using heated pressing dies to form a consolidated foil tab stack. Further, the method includes laser welding the consolidated foil tab stack to a terminal lead using a laser welder.

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Background/Summary

INTRODUCTION

[0001] The present disclosure relates to electrode stacks in a battery cell and more particularly to consolidating electrode tabs within the electrode stacks.

[0002] A battery cell, for example a prismatic battery cell, typically includes a plurality of electrode stacks. Each of the electrode stacks includes anode electrode tabs and cathode electrode tabs extending from the electrode stack. The electrode tabs of the electrode stacks are welded and connected to terminal leads or current collectors. The current collectors are metal plates that consolidate and transmit the electric current generated by the electrode stacks to the terminals on the battery cell. When welding the electrode tabs, surface contamination on the electrode tabs may cause gases to be released, which may form air cavities. These air cavities combined with tension caused by shrinkage during solidification may lead to detachment of electrode tabs at fusion boundaries.

[0003] While prior art methods and systems attempt to minimize detachment of electrode tabs at fusion boundaries and may achieve their particular purpose, a need still exists for a new and improved method for reducing detachment of the electrode tabs. Accordingly, a method that minimizes detachment of the electrode tabs is needed.

SUMMARY

[0004] According to several aspects of the present disclosure, a method for consolidating a foil tab stack is provided. The method includes combining foil tabs of an electrode stack in a battery cell to form a combined foil tab stack. Additionally, the method includes hot-pressing the combined foil tab stack using heated pressing dies to form a consolidated foil tab stack. Further, the method includes laser welding the consolidated foil tab stack to a terminal lead using a laser welder.

[0005] In accordance with another aspect of the disclosure, the method includes hot-pressing foil tabs formed from at least one of copper or aluminum.

[0006] In accordance with another aspect of the disclosure, the method includes heating the combined foil tab stack to a temperature between 350 Kelvin and 680 Kelvin and holding the temperature for between 5 seconds and 30 minutes.

[0007] In accordance with another aspect of the disclosure, the method includes using an induction-heated pressing die when hot-pressing the combined foil tab stack.

[0008] In accordance with another aspect of the disclosure, the method includes using a resistive-heated pressing die when hot-pressing the combined foil tab stack.

[0009] In accordance with another aspect of the disclosure, the method includes forming at least one indentation feature in the combined foil tab stack with the heated pressing dies when hot-pressing the combined foil tab stack.

[0010] In accordance with another aspect of the disclosure, the method includes forming at least one wave feature in the combined foil tab stack with the heated pressing dies when hot-pressing the combined foil tab stack.

[0011] In accordance with another aspect of the disclosure, the method includes forming at least one rib feature in the combined foil tab stack with the heated pressing dies when hot-pressing the combined foil tab stack.

[0012] In accordance with another aspect of the disclosure, the method includes using a high frequency oscillating laser when laser welding the combined foil tab stack.

[0013] In accordance with another aspect of the disclosure, the method further includes cooling at least one of a portion of a clamp or a subclamp configured to hold the electrode stack during hot-pressing the combined foil tab stack.

[0014] In accordance with another aspect of the disclosure, the method includes using a Peltier

plate when cooling at least the portion of the clamp.

[0015] In accordance with another aspect of the disclosure, the method includes using cooling channels in the subclamp when cooling at least the portion of the clamp.

[0016] According to several aspects of the present disclosure, a method for consolidating a foil tab stack is provided. The method includes combining foil tabs of an electrode stack in a battery cell to form a combined foil tab stack. The method includes clamping the combined foil tab stack to hold the combined foil tab stack in place. The electrode stack is supported by a clamp, and the combined foil tab stack is supported by a subclamp. The method includes hot-pressing the combined foil tab stack using heated pressing dies to form a consolidated foil tab stack. Moreover, the method includes laser welding the consolidated foil tab stack to a terminal lead using a laser welder.

[0017] In accordance with another aspect of the disclosure, the method includes heating the combined foil tab stack to a temperature between 350 Kelvin and 680 Kelvin and holding the temperature for between 5 seconds and 30 minutes when hot-pressing the combined foil tab stack.

[0018] In accordance with another aspect of the disclosure, the method includes using a pressing die that incorporates induction coils into a clamp that presses the combined foil tab stack when hot-pressing the combined foil tab stack. The induction coils heat the combined foil tab stack.

[0019] In accordance with another aspect of the disclosure, the method further includes cooling at least one of a portion of the clamp or the subclamp configured to hold the electrode stack during hot-pressing the combined foil tab stack.

[0020] According to several aspects of the present disclosure, a method for consolidating a foil tab stack is provided. The method includes combining foil tabs of an electrode stack in a battery cell to form a combined foil tab stack. The foil tabs are formed of copper or aluminum. The method includes clamping the combined foil tab stack to hold the foil tab stack in place. The electrode stack is supported by a clamp, and the combined foil tab stack is supported by a subclamp. The method includes hot-pressing the combined foil tab stack using heated pressing dies to form a consolidated foil tab stack. The method includes cooling at least a portion of the consolidated foil tab stack using a cooling system, where the cooling system is integrated into the subclamp. The method includes laser welding the consolidated foil tab stack to a terminal lead using a high frequency oscillating laser welder.

[0021] In accordance with another aspect of the disclosure, the method includes using a Peltier plate when cooling at least the portion of the clamp.

[0022] In accordance with another aspect of the disclosure, the method includes heating the combined foil tab stack to a temperature between 350 Kelvin and 680 Kelvin and holding the temperature for between 5 seconds and 30 minutes when hot-pressing the combined foil tab stack.

[0023] In accordance with another aspect of the disclosure, the method includes using cooling channels in the subclamp when cooling at least the portion of the clamp.

[0024] Further areas of applicability of the present disclosure will become apparent from the detailed description provided below. It should be understood that the detailed description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the disclosure.

[0025] The above features and advantages, and other features and advantages, of the presently disclosed system and method are readily apparent from the detailed description, including the claims, and examples when taken in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] The present disclosure will become more fully understood from the detailed description and the accompanying drawings, wherein:

[0027] FIG. 1 is a perspective view illustrating an example of a vehicle including a battery pack having a plurality of battery cells, in accordance with the present disclosure.

[0028] FIG. 2A is a perspective view illustrating a battery cell disposed within the battery pack shown in FIG. 1, where the battery cell includes at least one electrode stack having a plurality of electrode tabs welded to a current collector, in accordance with the present disclosure.

[0029] FIG. 2B is a partial cross-sectional view through the battery cell, as indicated in FIG. 2A, illustrating the layers in a single electrode stack and a set of anode electrode tabs and a set of cathode electrode tabs, in accordance with the present disclosure.

[0030] FIG. 3 is a flowchart illustrating a method for laser welding the anode and cathode electrode tabs to the current collectors using hot-pressing, in accordance with the present disclosure.

[0031] FIG. 4 is a perspective view illustrating an electrode stack and a combined foil tab stack shown in FIG. 2A, where a clamp and subclamp is supporting the electrode stack and combined foil tab stack, in accordance with the present disclosure.

[0032] FIG. 5 is a perspective view illustrating an electrode stack and a combined foil tab stack shown in FIG. 4, where a pressing die is hot-pressing the combined foil tab stack, in accordance with the present disclosure.

[0033] FIG. 6 is a perspective view illustrating the electrode stack and a consolidated foil tab stack resulting from hot-pressing the combined foil tab stack shown in FIG. 4, in accordance with the present disclosure.

[0034] FIG. 7 is an isometric perspective view illustrating the electrode stack and the consolidated foil tab stack shown in FIG. 4, in accordance with the present disclosure.

[0035] FIG. 8 is a perspective view illustrating a combined foil tab stack and pressing dies for hot-pressing the combined foil tab stack, in accordance with the present disclosure.

[0036] FIG. 9 is a perspective view illustrating a consolidated foil tab stack subsequent to hot-pressing the combined foil tab stack shown in FIG. 8, in accordance with the present disclosure.

[0037] FIG. 10 is a perspective view illustrating a combined foil tab stack and pressing dies for hot-pressing the combined foil tab stack, where the pressing dies are configured to create a feature in the resulting consolidated foil tab stack, in accordance with the present disclosure.

[0038] FIG. 11 is a perspective view illustrating a consolidated foil tab stack resulting from hot-pressing the combined foil tab stack shown in FIG. 10, where the features created in the consolidated foil tab stack include indentation features, in accordance with the present disclosure.

[0039] FIG. 12 is a perspective view illustrating a consolidated foil tab stack resulting from hot-pressing the combined foil tab stack shown in FIG. 10, where the features created in the consolidated foil tab stack include a wave feature, in accordance with the present disclosure.

[0040] FIG. 13 is a perspective view illustrating a consolidated foil tab stack resulting from hot-pressing the combined foil tab stack shown in FIG. 10, where the features created in the consolidated foil tab stack include rib features, in accordance with the present disclosure.

[0041] FIG. 14 is a perspective view illustrating a combined foil tab stack supported by a clamp and subclamp with a trimming feature for trimming an end of the combined foil tab stack, in accordance with the present disclosure.

[0042] FIG. 15 is a perspective view illustrating a consolidated foil tab stack trimmed with the trimming feature shown in FIG. 14, in accordance with the present disclosure.

[0043] FIG. 16 is a perspective view illustrating an electrode stack and combined foil tab stack supported by a clamp and a subclamp, where the clamp and the subclamp include a cooling system having cooling channels supplied by a coolant, in accordance with the present disclosure.

[0044] FIG. 17 is a perspective view illustrating an electrode stack and combined foil tab stack supported by a clamp and a subclamp, where the clamp and the subclamp include a cooling system having a Peltier plate, in accordance with the present disclosure.

[0045] FIG. 18 is a perspective view illustrating a consolidated foil tab stack welded to a terminal lead, where the resulting weld is a butt joint weld, in accordance with the present disclosure.

[0046] FIG. **19** is a perspective view illustrating a consolidated foil tab stack welded to a terminal lead, where the resulting weld is a lap joint weld with the weld location a distance from an end of the consolidated foil tab stack, in accordance with the present disclosure.

[0047] FIG. **20** is a perspective view illustrating a consolidated foil tab stack welded to a terminal lead, where the resulting weld is a lap joint weld with the weld location at end of the consolidated foil tab stack, in accordance with the present disclosure

[0048] FIG. **21** is a perspective view illustrating two consolidated foil tab stacks each supported by a clamp and a subclamp having Peltier plates, where one Peltier plate is heating the first consolidated foil tab stack and a second Peltier plate is cooling the second consolidated foil tab stack during a welding process, in accordance with the present disclosure.

DETAILED DESCRIPTION

[0049] Reference will now be made in detail to several examples of the disclosure that are illustrated in accompanying drawings. Whenever possible, the same or similar reference numerals are used in the drawings and the description to refer to the same or like parts or steps. The following description is merely exemplary in nature and is not intended to limit the present disclosure, application, or uses.

[0050] Referring to FIG. **1**, a perspective view of a vehicle **10** having a battery pack **12** is illustrated, in accordance with the present disclosure. The battery pack **12** is illustrated with an exemplary vehicle **10**. The vehicle **10** is an electric vehicle or hybrid vehicle having wheels **11** driven by electric motors/inverters **13**. The electric motors/inverters **13** receive power from the battery pack **12**. While the vehicle **10** is illustrated as a passenger road vehicle, it should be appreciated that the battery pack **12** may be used with various other types of vehicles. For example, the battery pack **12** may be used in nautical vehicles, such as boats, or aeronautical vehicles, such as drones or passenger airplanes. Moreover, the battery pack **12** may be used as a stationary power source separate and independent from a vehicle. Battery pack **12** includes a case **14** for supporting a plurality of battery cells **18**. The battery pack **12** may have fifty or more battery cells **18**.

[0051] Referring now to FIG. **2A**, a perspective view of a battery cell **18** disposed within the battery pack **12** shown in FIG. **1** is illustrated, in accordance with an aspect of the present disclosure. Each battery cell **18** has a housing or case (not shown) and a plurality of electrode stacks **22**. Each battery cell **18** may have hundreds of electrode stacks **22**. Each electrode stack **22** is connected to a current collector **24**, **26**. Moreover, electrode stacks **22** are placed in the housing and the housing is filled with a suitable electrolyte. For example, the electrolyte is a liquid solution of organic solvents and lithium salts. Current collectors **24**, **26** are thin metal plates or foils disposed on either side of the electrode stacks **22** typically having a thickness of between 0.4 and 1 millimeter. The current collectors may be made of copper or aluminum, for example. The current collectors **24**, **26** are attached to the electrode stacks **22** to transmit the electric current to an external circuit.

[0052] With additional reference to FIG. **2B**, a partial cross-sectional view through the battery cell **18**, as indicated in FIG. **2A**, illustrates the layers in a single electrode stack **22**, in accordance with the present disclosure. Each electrode stack **22** is comprised of a negative electrode or anode electrode **40** (a foil tab) extending from a first side of the electrode stack **22**, a separator **42**, and positive electrode or cathode electrode **44** (a foil tab) extending from a second side of the electrode stack **22**. The anode electrode **40** is generally a thin metal plate or foil that includes an anode electrode tab **46** (shown in FIG. **2A**) for providing an electrical connection between the anode electrode and current collector **24**. Similarly, cathode electrode **44** is a thin metal plate or foil that includes a cathode electrode tab **48** (shown in FIG. **2A**) for providing an electrical connection between the cathode electrode **44** and current collector **26**. Multiple cathode electrode tabs form a foil tab stack.

[0053] The anode electrode **40** and the anode electrode tab **46** are made, for example, of copper or other suitable material and typically coated with graphite or graphite/silicon or other carbon-based

materials or silicon oxide or lithiated silicon. The cathode electrode **44** and cathode electrode tab **48** are made, for example, of aluminum or other suitable material and typically coated with a metal oxide, such as Lithium cobalt oxide (LCO) or nickel-cobalt-aluminum (NCA) or lithium iron/manganese phosphate (LFP/LFMP) or Lithium manganese rich (LMR). The different metals (copper anode and aluminum cathode) of battery cell **18** produce a galvanic reaction in battery cell **18**. The copper, for example, of anode electrode **40**, and the aluminum, for example, of cathode electrode **44**, have different standard reduction potentials and are separated by one another by the separator **42**. The aluminum having the lower potential will oxidize and release electrons, while the copper having a higher potential will reduce and accept electrons. This process of releasing and accepting electrons generates an electric current that may be used to power devices. Multiple anode electrode tabs form a foil tab stack.

[0054] The separator **42** is generally a thin a porous membrane or layer of material that is positioned between the anode electrode **40** and the cathode electrode **44** and prevents the anode electrode **40** and cathode electrode **44** from touching and causing a short circuit. The separator **42** allows the lithium ions to pass through and complete the circuit. A composite material that is porous and chemically stable such as composites made with polyethylene (PE), polypropylene (PP) or other natural materials of the like may be used as the separator **42**. Moreover, inorganic nanoparticles such as TiO₂, SiO₂, Al₂O₃ and ZrO₂ may also be used to create coating composites for separator **42**. Separator **42** increases the internal resistance of battery cell **18** that reduces power output and efficiency of the battery. The internal resistance depends on the thickness porosity and composition of the separator **42**. The separator **42** is also selected to withstand high temperatures and manage thermal runaway preventing an uncontrollable rise in temperature due to exothermic reactions. Moreover, the separator **42** has a high melting point and a low shrinkage rate to avoid contact between the anode electrode **40** and cathode electrode **44**. The separator **42** has sufficient mechanical strength to resist puncture, tear, or deformation during fabrication and operation of cell assembly. The separator **42** also is dimensionally stable and flexible to conform to the shape of the electrodes and accommodate volume changes during cycling. The separator **42** is chemically inert and compatible with the electrolyte, electrodes and other cell components. Additionally, separator **42** has a low affinity for water or other impurities that can contaminate the electrolyte or cause corrosion of the electrodes.

[0055] With reference to FIG. **3**, a method **100** for consolidating a foil tab stack of the electrode stack **22** within the battery cell **18** is presented, in accordance with the present disclosure.

[0056] The method starts at block **102**. Block **102** depicts combining foil tabs **46**, **48** of the electrode stack **22** in the battery cell **18** to form a combined foil tab stack **52**. Combining the foil tabs **46**, **48** may include pre-assembling the electrode stack **22** in an upstream stacking process. For example, the anode electrodes **40**, the cathode electrodes **44**, and the separators may be deposited in a magazine (not shown) in a stacking wheel (not shown) and respectively arranged to form the electrode stack **22**. In this arrangement, the anode electrode tabs **46** extend from a first side of the electrode stack **22** and form a combined foil tab stack **52** comprising the anode electrode tabs **46**. The cathode electrode tabs **48** extend from another side of the first side of the electrode stack **22** or another portion of the first side of the electrode stack **22** (i.e. distant from the anode electrode tabs **46** and form a combined foil tab stack **52** comprising the cathode electrode tabs **48**. The combined foil tab stack **52** may include either anode electrode tabs **46** or cathode electrode tabs **48**. In one example, the combined foil tab stack **52** includes a stack of 50 electrode tabs, although the number of electrode tabs may vary (e.g., less than 25 electrode tabs to more than 100 electrode tabs. The method then may move to block **104**.

[0057] Block **104** depicts clamping the combined foil tab stack **52** in place using a clamp **54** or a subclamp **56**. Clamping the combined foil tab stack **52** may include using only a clamp **54** or both a clamp **54** and subclamp **56**. As used herein and shown in FIG. **4**, the clamp **54** is used to hold and support the electrode stack **22**, and the subclamp **56** is used to hold and support the combined foil

tab stack **52** in preparation for hot-pressing, although it should be appreciated that other configurations for the clamp **54** and the subclamp **56** may be used. Clamping can provide a joining force for additional combination of the combined foil tab stack **52**. Homogenous clamping of the combined foil tab stack **52** is beneficial due to low inherent stiffness of the electrode tabs in the combined foil tab stack **52** and residual stresses induced by rolling and winding processes during production and combination of the anode electrode tabs **46** and the cathode electrode tabs **48**. When the subclamp **56** is used, the subclamp **56** may contact only a portion of the combined foil tab stack **52** (e.g., a portion not intended for hot-pressing). In some instances, clamping the combined foil tab stack **52** may occur during a preassembly step and/or during an upstream stacking process.

Additionally, the clamp **54** and/or the subclamp **56** may be integrated with a pressing die **60**.

Further, the clamp **54** and/or the subclamp **56** may be heated using, for example, inductive heating (e.g., an induction-heated clamp and/or induction-heated subclamp) and/or resistive heating (e.g., a resistive-heated clamp and/or resistive-heated subclamp). A pressing die **60**, which incorporates induction coils into a non-conductive tooling, can press the combined foil tab stack **52**, and the induction coils heat the foils to engage diffusion bonding. The method then moves to block **106**.

[0058] Block **106** depicts hot-pressing the combined foil tab stack **52** using heated pressing dies **60** to form a consolidated foil tab stack **62**. Hot-pressing the combined foil tab stack **52** heats the combined foil tab stack **52** and presses the individual electrode tabs together. The hot-pressing step vaporizes surface contamination and moisture on the individual electrode tabs while minimizing air cavities within the consolidated foil tab stack **62**. Hot-pressing reduces porosity and detachment within the consolidated foil tab stack **62** in subsequent welding steps.

[0059] FIGS. **5**, **8**, **10**, **14-16**, and **21** illustrate a set of heated pressing dies **60** hot-pressing the combined foil tab stack **52** resulting in the consolidated foil tab stack **62**, such as the consolidated foil tab stack **62** illustrated in FIGS. **6** and **9**. The pressing dies **60** may include resistive-heated pressing dies **60**, induction-heated pressing dies **60**, and the like. In one specific example, a combined foil tab stack **52** including 50 tabs with an initial total thickness of 890 microns (μm) before a hot-pressing step is hot pressed resulting in a consolidated foil tab stack **62** having a resulting total thickness of about 812 μm . In this example, the difference in thickness is due to the hot-pressing step and elimination of surface contamination and air cavities that initially exist between the foil tabs **46**, **48**. In another example, hot-pressing the combined foil tab stack **52** may include heating the combined foil tab stack **52** to a temperature between 350 Kelvin (K) and 680 K and holding the temperature for between ten seconds and thirty minutes. It should be appreciated that a variety of temperatures and holding times may be used for hot-pressing. For example, the hot-pressing temperature may be from 300K or less to 900K or greater as long as a melting temperature of the combined foil tab stack **52** is not reached. The hot-pressing time may be from 5 seconds or less to 30 minutes or more.

[0060] Referring to FIGS. **10** through **13**, hot-pressing the combined foil tab stack **52** may include forming at least one feature in the combined foil tab stack **52**. FIG. **10** illustrates hot-pressing the combined foil tab stack **52** with heated pressing dies having features configured to impress and/or form the features. Referring to FIG. **11**, a consolidated foil tab stack **62** is shown having multiple indentation features **64** formed therein. The indentation features **64** may extend into and/or from the resulting consolidated foil tab stack **62**. The consolidated foil tab stack **62** may have one or more indentation features **64**. FIG. **12** illustrates a consolidated foil tab stack **62** having a wave feature **66** formed therein. The wave feature **66** may be formed on an outside surface and/or throughout the consolidated foil tab stack **62**. The wave feature **66** may facilitate improved gas ventilation during a subsequent welding process step. FIG. **13** illustrates a consolidated foil tab stack **62** having multiple rib features **68** formed therein. The rib features **68** may extend into and/or from the consolidated foil tab stack **62**. The consolidated foil tab stack **62** may have one or more rib features **68**. These features may serve to enhance foil tab consolidation, especially during transfer of the electrode stack **22** to subsequent process locations.

[0061] Additionally, hot-pressing the combined foil tab stack **52** may include trimming the combined foil tab stack **52** to allow a consistent clamping fit up. As shown in FIG. **14**, the heated pressing die **60** may include a trimming feature **70** that trims and removes a portion **72** of the combined foil tab stack **52** so that an end **74** of the resulting consolidated foil tab stack **62**, as shown in FIG. **15**, is even and has a consistent fit. The method **100** then may move to block **108**. [0062] Block **108** depicts cooling at least a portion of the subclamp **56** using a cooling system **76**. For example and referring to FIG. **16**, a subclamp **56** having cooling channels **78** may be used for the hot-pressing step depicted in block **106**. The cooling channels **78** are embedded and integral with the subclamp **56** and carry a coolant throughout at least a portion of the subclamp **56**. While the heated pressing dies **60** heat and press the combined foil tab stack **52**, the subclamp **56** cools a portion of the combined foil tab stack **52** and/or electrode stack **22** to avoid overheating the combined foil tab stack **52**.

[0063] In another example, and referring to FIG. **17**, cooling at least a portion of the subclamp **56** may include using a subclamp **56** including a Peltier plate **80**. A Peltier plate operates using the Peltier effect, which occurs when an electric current flows through a junction of two different materials. As a result, heat is transferred from one side of the plate (e.g., a cool side **82**) to another side of the plate (e.g., a hot side **84**). Depending on the direction of the current, the Peltier plate can either absorb heat (becoming cooler) or dissipate heat (becoming hotter). The Peltier plate **80** can be disposed integrally with the subclamp **56** or may be separate from the subclamp **56** and/or clamp **54**. The Peltier plate **80** can be arranged so that the hot side abuts and is proximate to the combined foil tab stack **52** and the cold side is away from and distal from the combined foil tab stack **52**. The method then moves to block **110**.

[0064] Block **110** depicts laser welding the consolidated foil tab stack **62** to a terminal lead **86** (e.g., a current collector) using a laser welder **88**. The laser welder **88** may include a high frequency oscillating laser welder. It will be appreciated that the laser welder **88** may include a variety of appropriate welders. In an example, laser welding the consolidated foil tab stack **62** may include using the laser welder **88** to high-beam oscillation weld the consolidated foil tab stack **62** to the terminal lead **86** with a vertical 8-shape oscillation pattern subsequent to a hot-pressing step.

[0065] Additionally, the consolidated foil tab stack **62** may be welded to the terminal lead **86** in a variety of configurations. For example, and as depicted in FIG. **18**, the consolidated foil tab stack **62** may be welded using the laser welder **88** to the terminal lead **86** in a butt joint configuration, where a resulting weld **90** is disposed at an end **92** of the consolidated foil tab stack **62** and the terminal lead **86**.

[0066] In the example depicted in FIG. **19**, the consolidated foil tab stack **62** is welded in a lap joint configuration, where the resulting weld **90** is disposed on a portion of the consolidated foil tab stack **62** with the consolidated foil tab stack **62** overlapping a portion of the terminal lead **86**.

[0067] In the example depicted in FIG. **20**, the consolidated foil tab stack **62** is welded in a lap joint configuration, where the resulting weld **90** is a fillet weld disposed on a portion of the consolidated foil tab stack **62** with the consolidated foil tab stack **62** overlapping a portion of the terminal lead **86**.

[0068] Referring to FIG. **21**, a heated pressing die **60A** may be used to heat a first consolidated foil tab stack **62A** located on a first side **98A** of the electrode stack **22**. A cooled pressing die **60B** may be used to cool a second consolidated foil tab stack **62B** located on a second side **98B** of the electrode stack **22**. This heating of the first consolidated foil tab stack **62A** and cooling of the second consolidated foil tab stack **62B** may be done simultaneously (e.g., during the welding step depicted in block **110**) or may be performed at separate times.

[0069] The present disclosure is advantageous and beneficial over prior art laser welding of electrode stack tabs. For example, hot-pressing the combined foil tab stack **52** serves to heat up the foil tabs **46**, **48** and vaporize surface contamination, often in the form of aluminum oxides, and eliminate air cavities. During subsequent welding steps, gases trapped in the resulting welds due to

the rapid solidification rate of aluminum alloys cause formation of pores. The pores, together with tension caused by solidification shrinkage, lead to detachment of foils at fusion boundaries. Using hot-pressing reduces porosity and detachment during subsequent welding.

[0070] This description is merely illustrative in nature and is in no way intended to limit the disclosure, its application, or uses. The broad teachings of the disclosure can be implemented in a variety of forms. Therefore, while this disclosure includes particular examples, the true scope of the disclosure should not be so limited since other modifications will become apparent upon a study of the drawings, the specification, and the following claims.

Claims

1. A method for consolidating a foil tab stack, comprising: combining foil tabs of an electrode stack in a battery cell to form a combined foil tab stack; hot-pressing the combined foil tab stack using heated pressing dies to form a consolidated foil tab stack; and laser welding the consolidated foil tab stack to a terminal lead using a laser welder.
2. The method of claim 1, wherein the foil tabs are formed from at least one of copper or aluminum.
3. The method of claim 1, wherein hot-pressing the combined foil tab stack includes heating the combined foil tab stack to a temperature between 350 Kelvin and 680 Kelvin and holding the temperature for between 5 seconds and 30 minutes.
4. The method of claim 1, wherein hot-pressing the combined foil tab stack includes using an induction-heated pressing die.
5. The method of claim 1, wherein hot-pressing the combined foil tab stack includes using a resistive-heated pressing die.
6. The method of claim 1, wherein hot-pressing the combined foil tab stack includes forming at least one indentation feature in the combined foil tab stack with the heated pressing dies.
7. The method of claim 1, wherein hot-pressing the combined foil tab stack includes forming at least one wave feature in the combined foil tab stack with the heated pressing dies.
8. The method of claim 1, wherein hot-pressing the combined foil tab stack includes forming at least one rib feature in the combined foil tab stack with the heated pressing dies.
9. The method of claim 1, wherein laser welding the combined foil tab stack includes using a high frequency oscillating laser.
10. The method of claim 1, further comprising: cooling at least one of a portion of a clamp or a subclamp configured to hold the electrode stack during hot-pressing the combined foil tab stack.
11. The method of claim 10, wherein cooling at least the portion of the clamp includes using a Peltier plate.
12. The method of claim 10, wherein cooling at least the portion of the clamp includes using cooling channels in the subclamp.
13. A method for consolidating a foil tab stack, comprising: combining foil tabs of an electrode stack in a battery cell to form a combined foil tab stack; clamping the combined foil tab stack to hold the combined foil tab stack in place, wherein the electrode stack is supported by a clamp, and wherein the combined foil tab stack is supported by a subclamp; hot-pressing the combined foil tab stack using heated pressing dies to form a consolidated foil tab stack; and laser welding the consolidated foil tab stack to a terminal lead using a laser welder.
14. The method of claim 13, wherein hot-pressing the combined foil tab stack includes heating the combined foil tab stack to a temperature between 350 Kelvin and 680 Kelvin and holding the temperature for between 5 seconds and 30 minutes.
15. The method of claim 13, wherein hot-pressing the combined foil tab stack includes using a pressing die that incorporates induction coils into a clamp that presses the combined foil tab stack, wherein the induction coils heat the combined foil tab stack.

16. The method of claim 13, further comprising: cooling at least one of a portion of the clamp or the subclamp configured to hold the electrode stack during hot-pressing the combined foil tab stack.

17. A method for consolidating a foil tab stack, comprising: combining foil tabs of an electrode stack in a battery cell to form a combined foil tab stack, wherein the foil tabs are formed of copper or aluminum; clamping the combined foil tab stack to hold the foil tab stack in place, wherein the electrode stack is supported by a clamp, and wherein the combined foil tab stack is supported by a subclamp; hot-pressing the combined foil tab stack using heated pressing dies to form a consolidated foil tab stack; cooling at least a portion of the consolidated foil tab stack using a cooling system, where the cooling system is integrated into the subclamp; and laser welding the consolidated foil tab stack to a terminal lead using a high frequency oscillating laser welder.

18. The method of claim 17, wherein cooling at least the portion of the clamp includes using a Peltier plate.

19. The method of claim 17, wherein hot-pressing the combined foil tab stack includes heating the combined foil tab stack to a temperature between 350 Kelvin and 680 Kelvin and holding the temperature for between 5 seconds and 30 minutes.

20. The method of claim 17, wherein cooling at least the portion of the clamp includes using cooling channels in the subclamp.
